

Sign-problem-free Bosonic Auxiliary Field Quantum Monte Carlo

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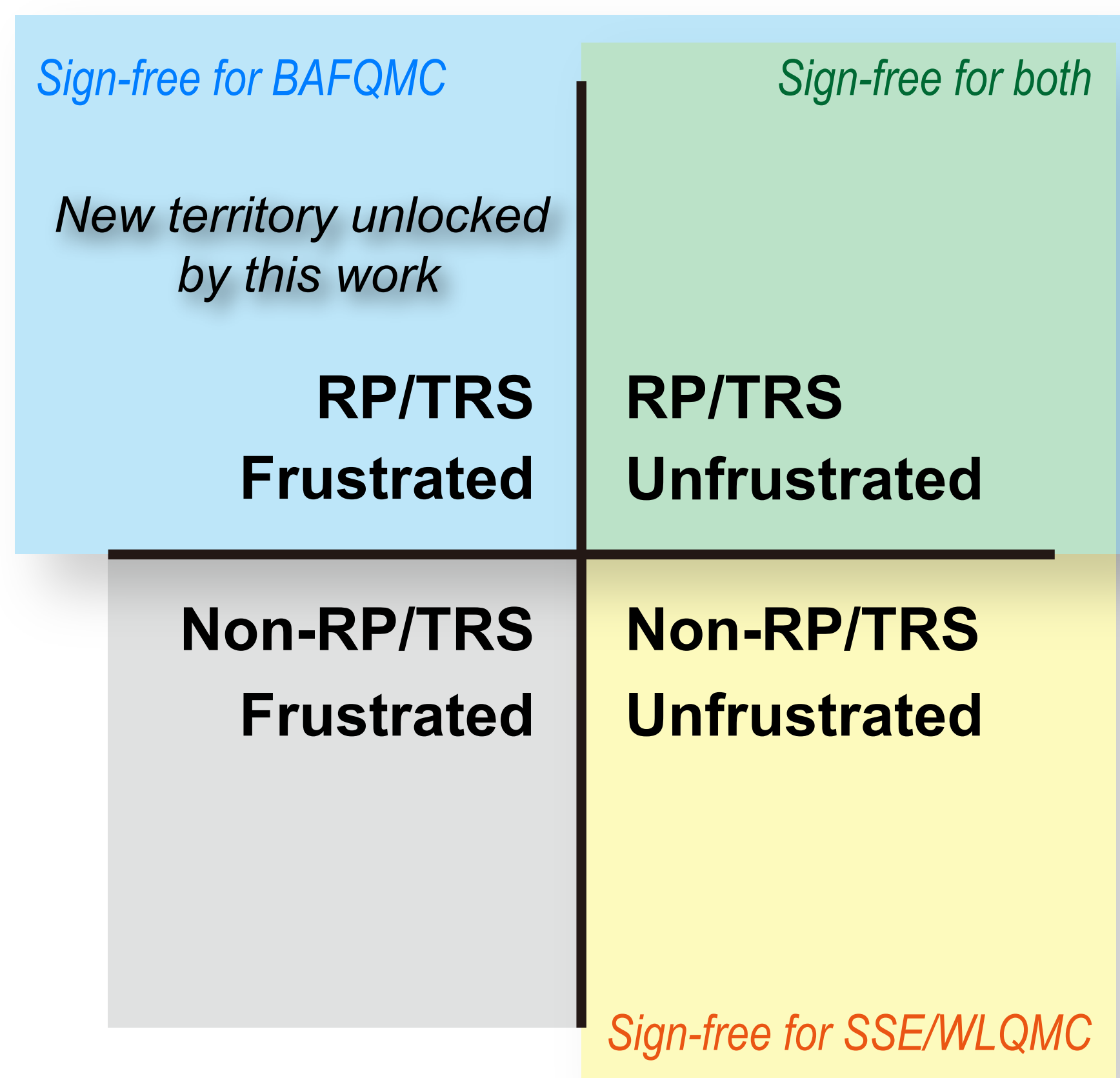
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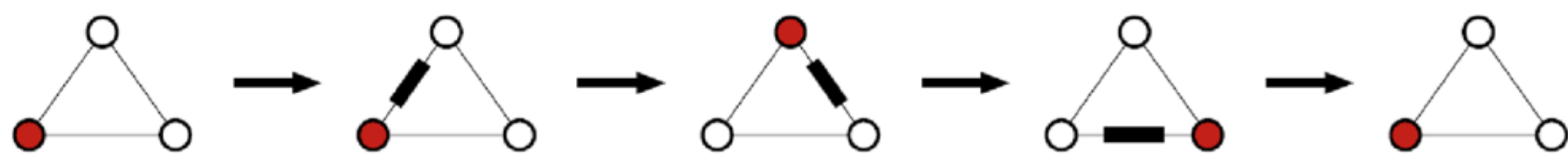
Sign Problem in Bosonic QMC

In this work, we propose a **sign-problem-free bosonic auxiliary-field quantum Monte Carlo (BAFQMC)** framework for generic interacting bosonic Hamiltonians. We identify and prove two sufficient conditions for sign-problem-free BAFQMC, namely **reflection positivity (RP)** and **time-reversal symmetry (TRS)**. Unlike in local-weight formulations, **geometric frustration** does NOT forbid sign-problem-free bosonic simulations once the problem is rewritten in auxiliary-field language, thereby opening a new exact-simulation territory **beyond world-line QMC and stochastic series expansion (SSE)**.



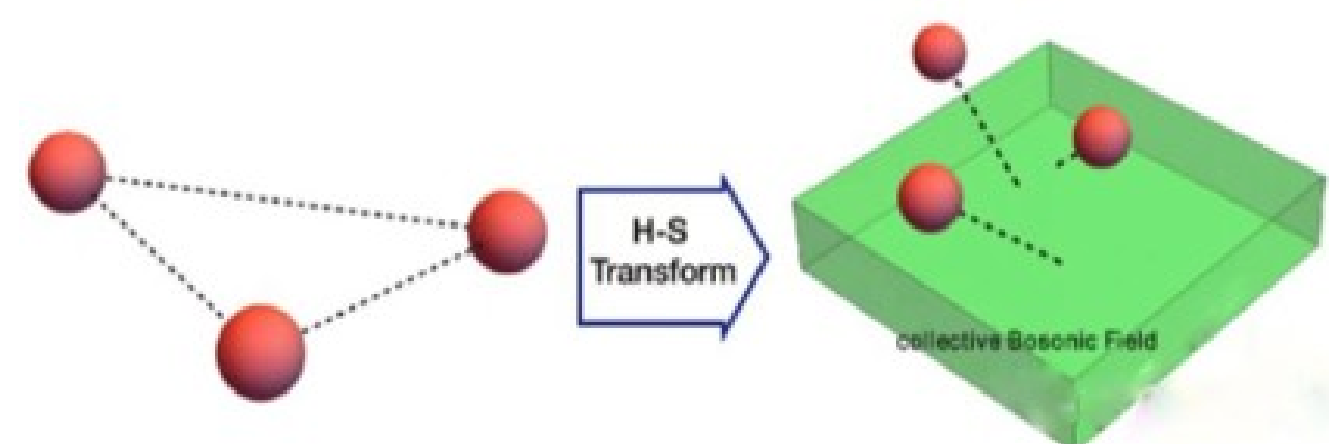
Sign problem in local-weight bosonic QMC (World-line / SSE)

Local amplitudes decide the sign. Frustrated hopping usually generates negative loops. For example, the antiferromagnetic Heisenberg interaction on a triangle: $W_{\text{loop}} = (J/2)^3 < 0$



Global-weight bosonic auxiliary-field QMC (BAFQMC)

Introduce an auxiliary field to decouple the interaction, trace out all local boson operators exactly, and then sample over the auxiliary-field ensemble—Positivity is controlled by the global symmetry of the decoupled Gaussian propagator.



Auxiliary Field Formalism for Bosonic Systems

Nambu basis $\hat{\chi} = (\hat{a}_1, \dots, \hat{a}_N, \hat{a}_1^\dagger, \dots, \hat{a}_N^\dagger)^\top$, one-body bosonic operator $\hat{h}(A, \xi) = -\frac{1}{2}\hat{\chi}^\top \Omega A \hat{\chi} + \xi$ encoded by the symplectic form $\Omega = \begin{pmatrix} 0 & I_N \\ -I_N & 0 \end{pmatrix}$, $A = \begin{pmatrix} K & \Delta \\ -\Delta & -K^\top \end{pmatrix}$.

Hubbard-Stratonovich decoupling

For a general interaction $\hat{H}_{\text{int}} = \sum_Y \lambda_Y \hat{O}_Y^2$, decouple:

$$e^{-\Delta\tau \lambda_Y \hat{O}_Y^2} = \int d\phi_Y p_Y(\phi_Y) e^{s_Y \phi_Y \hat{O}_Y} \Rightarrow Z = \int \mathcal{D}\phi P_{\text{aux}}(\phi) W(\phi).$$

Bosonic Gaussian trace formula

$$W(\phi) = \text{Tr} \left[e^{\hat{h}_1(\phi_1)} \dots e^{\hat{h}_L(\phi_L)} \right] = e^{\sum \xi_\ell(\phi_\ell)} \det(I_{2N} - \mathcal{M}(\phi))^{-1/2}, \quad \mathcal{M}(\phi) = e^{A_1} \dots e^{A_L}.$$

Importance sampling and local updates

Considering a shift $e^{A_\ell} = (I_{2N} + \Delta_\ell) e^{A_\ell}$, $\delta\xi_\ell = \xi'_\ell - \xi_\ell$, the Gaussian weight ratio:

$$\frac{W(\phi')}{W(\phi)} = e^{\delta\xi_\ell} \det[I_{2N} + \Delta_\ell (I_{2N} - G_\ell \Omega^{-1})]^{-1/2},$$

where the Green's function is $G(\phi) = (I_{2N} - \mathcal{M}(\phi))^{-1} \Omega$.

Two Sufficient Conditions for Nonnegative Weights

Reflection Positivity (RP)

Reflection map $\theta : i \rightarrow -i, \hat{b}_i \rightarrow \hat{c}_i, \hat{c}_i \rightarrow \hat{b}_i$. Reflective positivity at a heuristic level: $\text{Tr} \left[e^{\hat{h} + \theta(\hat{h})} \right] = \text{Tr} \left[e^{\hat{h}} \theta(e^{\hat{h}}) \right] > 0$. For examples, a useful reflection-positive form:

$$\hat{h} = \hat{\chi}^\top \begin{pmatrix} D & F \\ F^\top & \bar{D} \end{pmatrix} \hat{\chi} + \xi, \quad \hat{\chi} = (\hat{\mathbf{b}}, \hat{\mathbf{b}}^\dagger, \hat{\mathbf{c}}, \hat{\mathbf{c}}^\dagger)^\top, \quad F = F^\dagger \geq 0.$$

Time-Reversal Symmetry (TRS)

Time-reversal map: $\Theta : i \rightarrow -i, \hat{b}_i \rightarrow i\hat{c}_i, \hat{c}_i \rightarrow -i\hat{b}_i$. Time-reversal symmetry: $\Theta \hat{h} \Theta^{-1} = \hat{h}$. All the eigenvalues of $\mathcal{B} \equiv e^{B_1} \dots e^{B_L}$ appear in complex conjugate pairs, due to $\Theta^2 = -1$ Kramers degeneracy, so $\det(I_N - \mathcal{B})^{-1} > 0$. A useful sign-free block form:

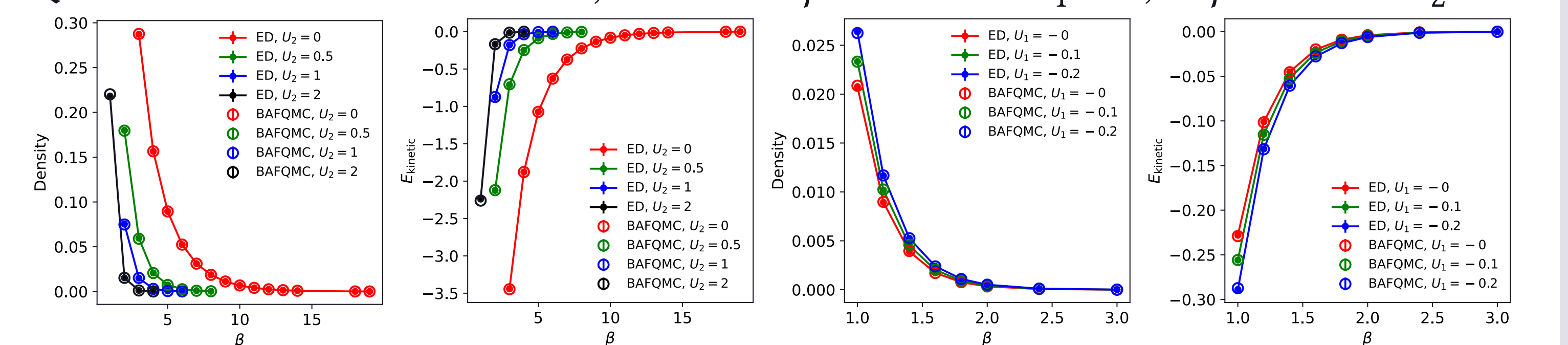
$$\hat{h} = \hat{\mathbf{a}}^\dagger \begin{pmatrix} D & F \\ -F & \bar{D} \end{pmatrix} \hat{\mathbf{a}} + \xi, \quad \hat{\mathbf{a}} = (\hat{\mathbf{b}}, \hat{\mathbf{c}})^\top.$$

Triangular-Lattice Benchmarks

Two-flavor frustrated Bose-Hubbard models

$$\hat{H} = t \sum_{\langle i,j \rangle} (\hat{b}_i^\dagger \hat{b}_j + \hat{c}_i^\dagger \hat{c}_j + \text{h.c.}) + U_1 \sum_i (\hat{b}_i^\dagger \hat{b}_i + \hat{c}_i^\dagger \hat{c}_i)^2 + U_2 \sum_i (\hat{b}_i^\dagger \hat{b}_i - \hat{c}_i^\dagger \hat{c}_i)^2, \quad U_1 \leq 0, U_2 \geq 0.$$

QMC vs ED results at 2×3 sites, with fixed $\mu = -2.5$ for $U_1 = 0$; or $\mu = -5$ for $U_2 = 1$



One More Thing

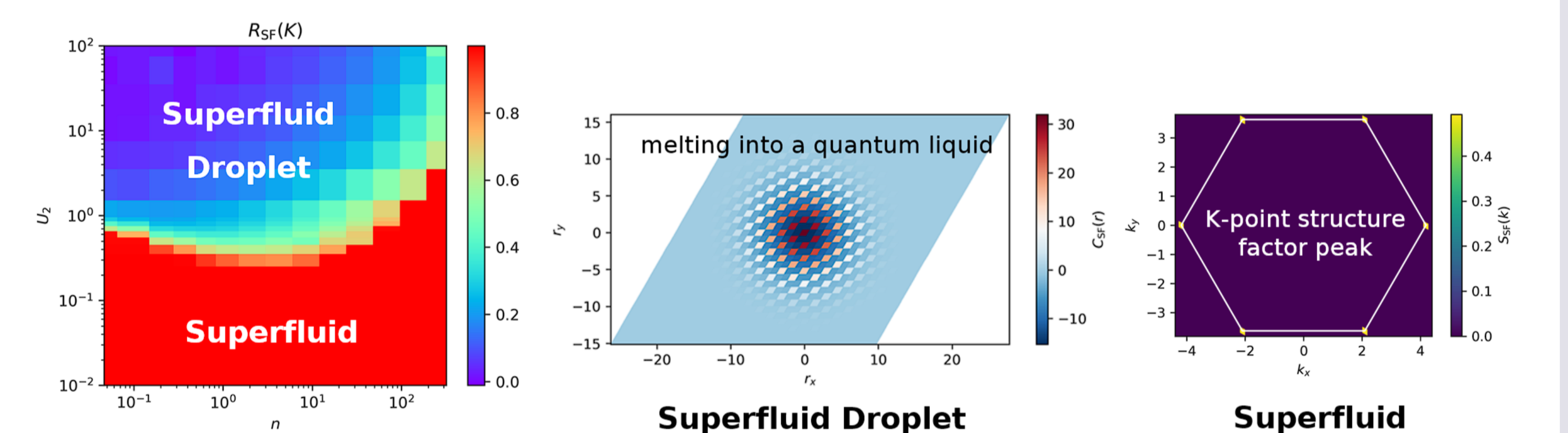
sign-problem-free Bosonic projector QMC [$O(N^2 N_\tau)$ complexity]

$$\text{Rank-1 trial state: } |\Phi_T\rangle = \frac{1}{\sqrt{N_b!}} \left(\sum_i \hat{b}_i^\dagger P_i \right)^{N_b} |0\rangle \otimes \frac{1}{\sqrt{N_b!}} \left(\sum_i \hat{c}_i^\dagger P_i^* \right)^{N_b} |0\rangle$$

$$\text{Projector: } \hat{U}_b(\tau_2, \tau_1) = e^{-\hat{b}^\dagger B(\tau_2) \hat{b}} \dots e^{-\hat{b}^\dagger B(\tau_1) \hat{b}}, \quad \mathcal{B}(\tau_2, \tau_1) = e^{-B(\tau_2)} \dots e^{-B(\tau_1)}$$

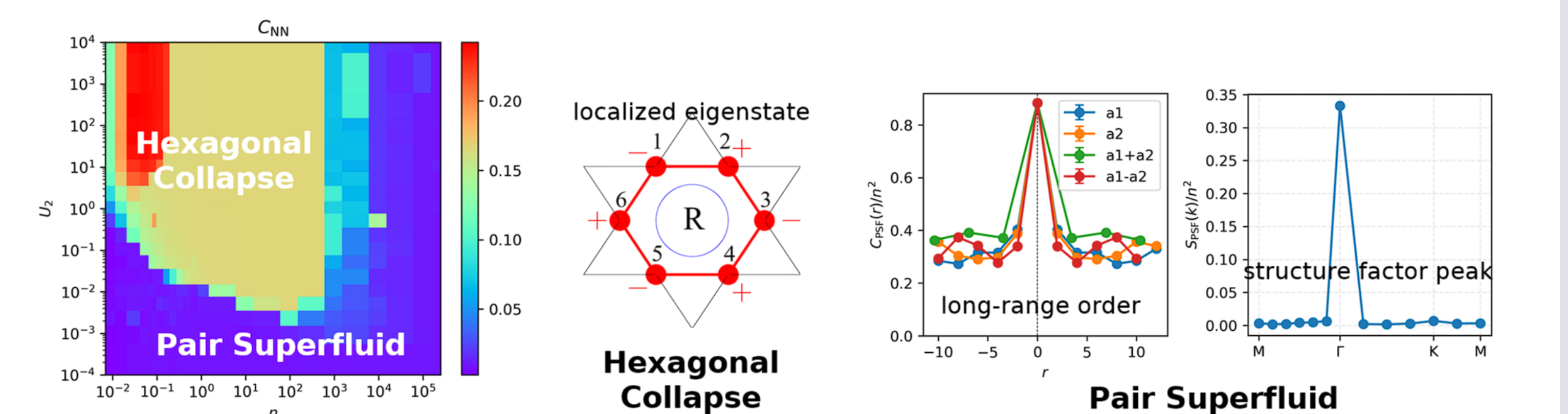
$$\text{Sampling weight: } W(\phi) = e^{-\frac{1}{2}|\phi|^2} \left([P^\dagger \mathcal{B}(2\theta, 0) P]^{N_b} \right) \left([P^\dagger \mathcal{B}(2\theta, 0) P]^{N_b} \right)^* \geq 0$$

Solving triangular lattice frustrated Bose-Hubbard model (up to 60×60 sites)

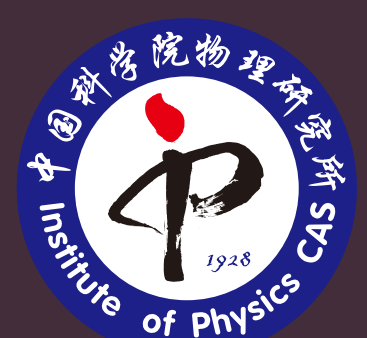


Superfluid droplet: a spatially self-bound quantum liquid without long-range order.

Solving Kagome lattice frustrated Bose-Hubbard model (up to $36 \times 36 \times 3$)



Pair superfluid: $\langle \hat{b}_i^\dagger \hat{b}_j \rangle = 0$ but $\langle \hat{b}_i^\dagger \hat{c}_i^\dagger \hat{b}_j \hat{c}_j \rangle \neq 0 \Rightarrow$ fermionic charge-4e SC



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Preprint: Coming soon!

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Reference

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